

3E3 Modelling Risk Coursework

Integrated Risk Analysis for GreenCo Energy

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1 Introduction

GreenCo Energy is developing a coastal renewable energy hub integrating offshore wind generation, on-site hydrogen production via electrolysis, and a heavy goods vehicle (HGV) refuelling station. The project faces four interlinked sources of uncertainty: **demand volatility** (stochastic monthly hydrogen demand), **production variability** (electrolyser availability), **congestion risk** (queueing at the refuelling station), and **long-term investment uncertainty** (demand growth trajectory and competitor entry).

This report develops an integrated risk analysis. Four models are applied, each addressing a different risk source: (1) *Newsvendor model* for the monthly production decision under demand uncertainty; (2) *M/M/s queueing model* for refuelling station congestion; (3) *Decision tree* for multi-stage strategy evaluation under scenario uncertainty; (4) *Monte Carlo simulation* for full NPV distributions and downside risk measurement. The objective is to maximise expected NPV while explicitly assessing downside risk exposure and service-level compliance.

2 Problem Setup

2.1 Analytical Models

We use the newsvendor inventory production model for computing the expected profit $\mathbb{E}[\Pi_t]$ and optimal production quantity at year t , Q_t^* . We also use a M/M/s queueing model to model refuelling service. Our parameters for these models, and demand growth models, are shown in Table 1. Exact details are explained in section 3.

Table 1: Model parameters.

Symbol	Parameter	Value
<i>Demand & Pricing</i>		
μ_0	Initial mean monthly demand	1000 t
σ_0	Initial demand std. deviation	150 t
CV	Coefficient of variation (σ_t / μ_t)	0.15
μ_t	Year- t mean monthly demand	Eqn (1)
σ_t	Year- t standard deviation of monthly demand	Eqn (1)
p	Selling price	£4000/t
c_p	Production cost	£2500/t
c_s	Spot market purchase price	£3200/t
c_h	Storage/handling cost (unsold)	£300/t
<i>Production Capacity</i>		
K_{nom}	Nominal electrolyser capacity	40 t/day
a	Operational availability	85%
n_d	Days per month	30
<i>Refuelling Station</i>		
λ	HGV arrival rate	10 vehicles/hr
$1/\mu$	Mean refuelling time	12 min
s_0	Initial number of bays	3
W_q^{max}	Maximum average queue wait	5 min
<i>Growth & Discounting</i>		
p_H	Probability of high growth	0.45
g_H, g_L	Annual growth rates (mean demand)	10%, 3%
p_C	Probability of competitor entry (high growth, Y3)	0.40
δ	Demand reduction from competitor	15%
r	Discount rate	7%/yr

2.2 Demand Growth and Scenarios

Monthly hydrogen demand in year t : $D_t \sim \mathcal{N}(\mu_t, \sigma_t^2)$, with constant $CV = \sigma_t/\mu_t = 0.15$, and annual growth rate of mean demand g gives:

$$\mu_t = \mu_0(1 + g)^t, \quad \sigma_t = CV \cdot \mu_t \quad (1)$$

With competitor entry (high growth scenario, $t \geq 3$): $\mu_t^{\text{comp}} = 0.85 \mu_t$. This gives three terminal scenarios:

Table 2: Scenario probabilities and monthly demand (μ_t in t/month, σ_t in t/month).

Yr	High, no comp. ($p = 0.27$)		High + comp. ($p = 0.18$)		Low growth ($p = 0.55$)	
	μ_t	σ_t	μ_t	σ_t	μ_t	σ_t
0	1000	150	1000	150	1000	150
1	1100	165	1100	165	1030	155
2	1210	182	1210	182	1061	159
3	1331	200	1131	170	1093	164
4	1464	220	1245	187	1126	169

2.3 Investment Strategies

GreenCo must choose one of three strategies at Year 0 to scale its hydrogen infrastructure over a 4-year horizon. Each strategy modifies a different subset of the newsvendor parameters ($c_s, \sigma_t, \bar{Q}$) and the queueing parameter s , creating distinct risk–return profiles.

Strategy 1 (Immediate Scale-Up). Invest £54M upfront (£48M electrolyser expansion + £6M for two additional refuelling bays) to increase capacity from $K_{\text{nom}} = 40$ to 65 t/day at 95% availability, giving $\bar{Q}_{S1} = 1852.5$ t/mo, and $s = 5$ bays. No changes to demand uncertainty or spot pricing. The bet is that high capacity eliminates spot procurement: since $\bar{Q}_{S1} \gg \mu_t$ for all t , the capacity constraint in the newsvendor never binds. However, the risk is overcapacity under the low growth scenario — demand utilisation $\mu_t/\bar{Q}_{S1}$ falls below 65% from Y2, triggering a £150/t scale-inefficiency penalty that inflates c_p to £2650/t.

Strategy 2 (Flexibility and Risk Mitigation). Invest £20M upfront in modular storage, demand-smoothing contracts, and digital forecasting. This does not expand capacity ($\bar{Q}, s$ are unaffected), but instead reduces two key newsvendor cost drivers: demand volatility drops by 40% ($\sigma_t \rightarrow 0.6\sigma_t = 0.09\mu_t$), and pre-negotiated contingent contracts lower spot procurement cost to $c_s = £2900/\text{t}$. The reduced c_s shifts the critical ratio from $F(Q^*) = 0.200$ to $F(Q^*) = 0.125$, pushing the optimal production quantity further below the mean; simultaneously, the tighter demand distribution ($\sigma \downarrow$) reduces both ELS and ELOI, cutting procurement and storage waste. Under high growth, capacity eventually binds ($Q_t^* > \bar{Q}$ from Y3), forcing systematic spot reliance.

Strategy 3 (Staged Expansion). Invest £8M upfront in expandable foundations and grid upgrades, preserving the option to scale later. This is a *real-options* structure: at the end of Y1, after observing demand growth, GreenCo chooses one of two branches:

- *High growth observed* $\rightarrow$ expand nominal capacity $K_{\text{nom}} = 65$ t/day at $a = 95\%$ availability for £42M (discounted by the pre-installed foundations), matching S1 specifications from Y2 onward.
- *Low growth observed* $\rightarrow$ invest £15M in flexibility measures (a reduced version of S2: same σ reduction and $c_s = £2900/\text{t}$), matching S2 specifications from Y2 onward.

The cost of deferral is a 10% realised revenue reduction in Y1–Y2 under both growth paths. Until the Y1 decision, capacity and parameters remain at baseline.

Table 3 summarises the parameter modifications. All unmodified parameters retain their base-case values from Table 1.

Table 3: Strategy-specific parameter modifications (base: $\bar{Q} = 1020$ t/mo, $s = 3$ bays, $c_s = \text{£}3200/\text{t}$, $\sigma_t = 0.15\mu_t$).

Parameter	Strategy 1	Strategy 2	Strategy 3
I_0 (Year 0)	£54M	£20M	£8M
I_1 (Year 1)	—	—	£42M (H) / £15M (L)
Capacity $\bar{Q}$	1852.5 t/mo	1020 t/mo	1020 $\rightarrow$ 1852.5 (H branch)
Availability	95%	85%	85% $\rightarrow$ 95% (H branch)
Bays s	5	3	3
σ_t	$0.15\mu_t$	$0.09\mu_t$	$0.15\mu_t$
c_s	£3200/t	£2900/t	£3200/t
Revenue	100%	100%	90% in Y1–Y2
Low-growth c_p penalty	+£150/t from Y2	—	—

3 Operational Risk Analysis

3.1 Monthly Production Decision: Newsvendor Model

Each month GreenCo chooses production quantity $Q \in [0, \bar{Q}]$ before demand $D \sim \mathcal{N}(\mu, \sigma^2)$ is realised. The effective monthly capacity $\bar{Q}$ is the product of nominal electrolyser throughput, operational availability, and operating days:

$$\bar{Q} = K_{\text{nom}} \cdot a \cdot n_d = 40 \times 0.85 \times 30 = 1020 \text{ t/mo} \quad (2)$$

All demand is contractually met; shortfalls are covered by spot procurement. Monthly profit:

$$\Pi(Q, D) = \underbrace{pD}_{\text{revenue}} - \underbrace{c_p Q}_{\text{production}} - \underbrace{c_s \max(D - Q, 0)}_{\text{spot procurement}} - \underbrace{c_h \max(Q - D, 0)}_{\text{storage}} \quad (3)$$

Taking expectations over demand D , we define *expected lost sales* (demand unmet by own production) and *expected leftover inventory* (unsold overproduction):

$$\text{ELS}(Q) = \mathbb{E}_D[\max(D - Q, 0)], \quad \text{ELOI}(Q) = \mathbb{E}_D[\max(Q - D, 0)] \quad (4)$$

Since revenue pD is independent of Q (contractual obligation), expected profit is:

$$\mathbb{E}_D[\Pi(Q)] = p\mu - c_p Q - c_s \text{ELS}(Q) - c_h \text{ELOI}(Q) \quad (5)$$

3.1.1 Optimal Production Quantity

Note $\frac{d}{dQ} \text{ELS} = -(1 - F(Q))$ and $\frac{d}{dQ} \text{ELOI} = F(Q)$, where F is the CDF of demand D . Setting $d\mathbb{E}[\Pi]/dQ = 0$:

$$\frac{d\mathbb{E}[\Pi]}{dQ} = -c_p + c_s(1 - F(Q)) - c_h F(Q) = 0 \implies \boxed{F(Q^*) = \frac{c_s - c_p}{c_s + c_h} = \frac{c_u}{c_u + c_o}} \quad (6)$$

This is the *newsvendor critical ratio*. In the standard newsvendor with salvage value s , underage and overage costs are $c_u = p - c$ and $c_o = c - s$. Here, because revenue is Q -independent, underage costs arise from spot procurement and overage costs from production plus storage with zero salvage (effective $s_{\text{eff}} = -c_h$, giving $c_o = c_p + c_h$):

$$c_u = c_s - c_p = \text{£}700/\text{t} \quad (\text{extra cost of spot vs own production}) \quad (7)$$

$$c_o = c_p + c_h = \text{£}2800/\text{t} \quad (\text{production + storage on unsold units}) \quad (8)$$

The ratio $c_o/c_u = 4$ is strongly asymmetric: overproducing is four times more expensive per tonne than underproducing. Therefore $F(Q^*) = c_u/(c_u + c_o) = 700/3500 = 0.20$. For normal demand $Q^* = \mu + z^*\sigma$ where $z^* = \Phi^{-1}(0.20) = -0.8416$. The firm optimally produces *well below* mean demand, covering the shortfall from the relatively cheap spot market. Optimal production quantity at $t = 0$ is:

$$Q_0^* = \mu_0 + z^*\sigma_0 = 1000 + (-0.8416)(150) \approx 874 \text{ t/month} < \bar{Q} = 1020 \quad (9)$$

3.1.2 Expected Profit Components

For normal demand, the standard normal loss function $L(z) = \phi(z) - z[1 - \Phi(z)]$ gives closed-form expressions for ELS and ELOI. With $z^* = -0.8416$ and $L(-0.8416) = 0.953$:

$$\text{ELS} = \sigma_0 L(z^*) = 150 \times 0.953 = 143.0 \text{ t} \quad (\text{spot procurement volume}) \quad (10)$$

$$\text{ELOI} = \sigma_0 [z^* + L(z^*)] = 150 \times 0.111 = 16.7 \text{ t} \quad (\text{unsold overproduction}) \quad (11)$$

Using Eqn (5), our expected profit at $t = 0$ is:

$$\mathbb{E}[\Pi_0] = p\mu - c_p Q^* - c_s \cdot \text{ELS} - c_h \cdot \text{ELOI} = \text{£}1353020 \approx \text{£}1.35\text{M/month} \quad (12)$$

giving an annual baseline of **£16.24M**.

3.1.3 Evolution Under Demand Growth and Capacity Constraints

The unconstrained optimum is $Q_t^* = \mu_t(1 + \text{CV} \cdot z^*) = 0.874 \mu_t$. Under high growth, $0.874 \mu_2 = 1058 > \bar{Q} = 1020$, so *the capacity constraint binds from Year 2 onward* under Strategies 2 and 3 (before expansion). Once binding, $Q_t^* = \bar{Q}$ and ELS grows rapidly as demand outpaces production. This is the principal mechanism forcing emergency spot procurement and limiting profits under Strategies 2 and 3 with high growth.

Table 4 summarises expected annual profit under each strategy, scenario, and year.

Table 4: Expected annual profit $\mathbb{E}[\Pi_t^{(k,j)}]$ (£M) by strategy, scenario, and year.

Strategy	Scenario	Y1	Y2	Y3	Y4
S1	High, no comp.	17.86	19.65	21.61	23.77
	High + comp.	17.86	19.65	18.37	20.21
	Low growth	16.72	15.58	16.05	16.53
S2	High, no comp.	19.02	20.80	22.46	24.22
	High + comp.	19.02	20.80	19.56	21.29
	Low growth	17.81	18.34	18.89	19.46
S3	High, no comp.	12.58	13.84	21.61	23.77
	High + comp.	12.58	13.84	18.37	20.21
	Low growth	11.78	13.25	18.89	19.46

Key observations from Table 4: (i) S2 consistently produces higher annual profit than S1 in every scenario and year, despite identical revenue—the reduced spot price (£2900/t) and tighter demand distribution (lower σ) jointly reduce procurement and storage costs; (ii) S1’s low-growth profits fall sharply in Y2–Y4 due to the £150/t scale-inefficiency penalty; (iii) S3’s Y1–Y2 profits are suppressed by the 10% revenue reduction, recovering only from Y3 once the Year-1 investment takes effect.

3.2 Refuelling Station Congestion: M/M/s Queue

HGVs arrive as a Poisson process ($\lambda = 10/\text{hr}$), refuelling times are i.i.d. $\text{Exp}(\mu)$ with mean $1/\mu = 12 \text{ min}$ ($\mu = 5/\text{hr}$ per bay), and there are $s_0 = 3$ bays in parallel (FIFO, infinite buffer). This is an M/M/3 queue.

3.2.1 Baseline Performance

With $a = \lambda/\mu = 2$ and $\rho = \lambda/(s\mu) = 10/15 = 2/3 < 1$ (stable):

$$P_0 = \left[\sum_{n=0}^2 \frac{2^n}{n!} + \frac{2^3}{3!} \cdot \frac{1}{1 - 2/3} \right]^{-1} = [1 + 2 + 2 + 4]^{-1} = \frac{1}{9} \quad (13)$$

$$L_q = \frac{a^s \rho}{s!(1 - \rho)^2} P_0 = \frac{8 \times (2/3)}{6 \times (1/3)^2} \times \frac{1}{9} = \frac{8}{9} \approx 0.889 \text{ vehicles} \quad (14)$$

$$W_q = \frac{L_q}{\lambda} = \frac{0.889}{10} = 0.0889 \text{ hr} = \mathbf{5.33 \text{ min}} \quad (15)$$

By Little’s Law: $L = \lambda W = 2.89$ vehicles in system, $W = W_q + 1/\mu = 17.3 \text{ min}$ total sojourn.

Finding: Even at baseline, $W_q = 5.33 \text{ min} > W_q^{\max} = 5 \text{ min}$. GreenCo is *already* non-compliant with its service-level agreement before any demand growth. The maximum arrival rate compatible with compliance is $\lambda_{s=3}^{\max} = 9.84/\text{hr}$, i.e. just below the current rate of 10/hr.

3.2.2 Impact of Demand Growth

Assuming $\lambda_t = \lambda_0(1 + g)^t$, Table 5 shows how W_q evolves. Under either growth scenario with $s = 3$, the wait time violates the 5-minute constraint from Year 0 and grows catastrophically—reaching 159.6 min by Y4 under high growth as $\rho \rightarrow 1$.

With $s = 5$ bays (Strategy 1), $\rho = 0.40$ at baseline and W_q remains below 1.3 min across the entire 4-year horizon under any growth scenario. The maximum compliant λ for $s = 5$ is 19.13/hr, far exceeding any projected arrival rate.

Table 5: W_q (minutes) under demand growth. Values marked * exceed the 5-min target.

Year	High growth			Low growth		
	λ (/hr)	$W_q, s=3$	$W_q, s=5$	λ (/hr)	$W_q, s=3$	$W_q, s=5$
0	10.00	5.33*	0.24	10.00	5.33*	0.24
1	11.00	8.13*	0.36	10.30	6.04*	0.27
2	12.10	13.62*	0.54	10.61	6.87*	0.31
3	13.31	28.22*	0.82	10.93	7.88*	0.35
4	14.64	159.63*	1.27	11.26	9.10*	0.40

A fourth bay ($s = 4$) raises the compliant threshold to 14.44/hr, sufficient under low growth across all years but insufficient under high growth by Y4 ($\lambda_4 = 14.64/\text{hr}$). **Service-level compliance under any growth scenario requires adding at least two bays ($s = 5$), regardless of the chosen investment strategy.** This is a critical operational recommendation that Strategy 2 and 3 fail to address.

4 Strategic Investment Analysis

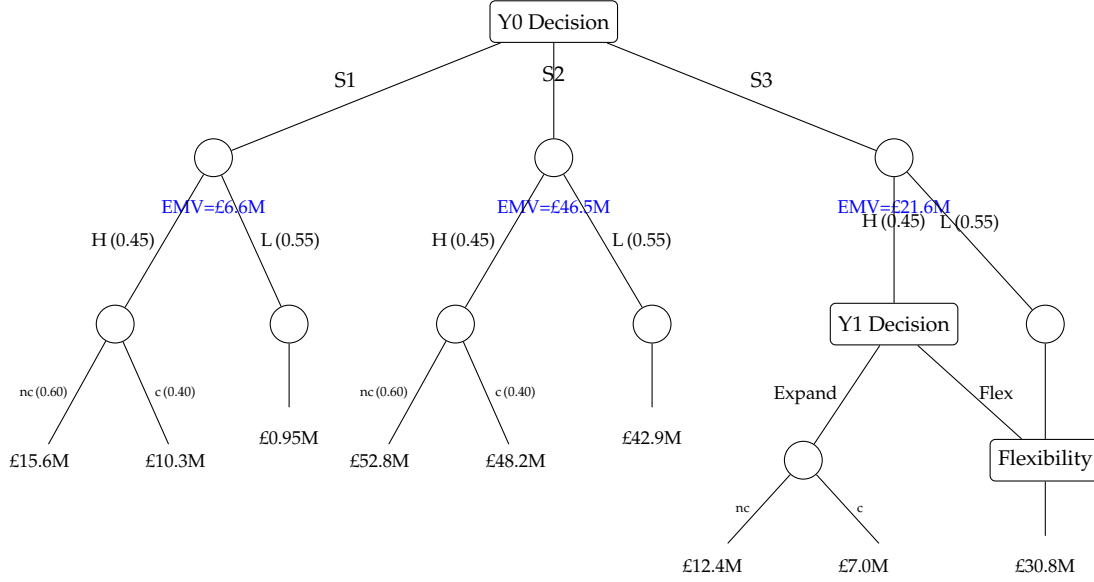
4.1 Decision Tree and NPV Formulation

The three strategies are evaluated over a 4-year horizon using a multi-stage decision tree (Figure 1). Strategy 3 has a real-options structure: a second decision node at the end of Y1 lets GreenCo select the expansion or flexibility branch conditional on observed growth. This is solved by *backward induction*.

The NPV for strategy k under scenario j is:

$$\text{NPV}^{(k,j)} = -I_0^{(k)} + \sum_{t=1}^4 \frac{\mathbb{E}[\Pi_t^{(k,j)}]}{(1+r)^t} - \sum_{t \geq 1} \frac{I_t^{(k,j)}}{(1+r)^t} \quad (16)$$

where $I_1^{(k,j)}$ is discounted at $t = 1$ (paid at end of Year 1). Annual profits $\mathbb{E}[\Pi_t^{(k,j)}]$ are from Table 4.



nc = no competitor entry; c = competitor enters Y3; S3 Y1 decision conditional on observed growth.

Figure 1: Simplified decision tree. Terminal values are NPVs under each scenario. H branch of S3 shows two terminals (no comp./comp.). EMV computed by backward induction.

4.2 NPV Results

Table 6: NPV by strategy and scenario (£M), with expected NPV.

Scenario	S1	S2	S3
High, no comp. ($p = 0.27$)	15.63	52.75	12.37
High + comp. ($p = 0.18$)	10.26	48.15	7.00
Low growth ($p = 0.55$)	0.95	42.93	30.83
E[NPV]	6.59	46.52	21.55

Strategy 2 dominates by a large margin in every scenario. Its E[NPV] of £46.5M exceeds S1 by £39.9M and S3 by £25.0M. The mechanism is threefold: (i) much lower capex (£20M vs £54M/£50M equivalent) leaves less invested capital to recover; (ii) reduced c_s from £3200 to £2900/t lowers the underage penalty, shifting the critical ratio from 0.200 to 0.125 and reducing expected spot procurement costs; (iii) reduced demand volatility ($\sigma \rightarrow 0.6\sigma$) lowers both expected leftover and expected shortfall, simultaneously reducing storage and procurement waste.

S1 performs poorly under low growth (NPV = £0.95M) due to the combination of high capex and the scale-inefficiency cost penalty. S3's real-options value—the ability to defer the large expansion cost—is insufficient to compensate for the 10% Y1–Y2 revenue penalty and the fact that Strategy 2 is already cheaper and more profitable in every state.

5 Risk Assessment

5.1 Monte Carlo Simulation Results

For each of $N = 50000$ simulated paths: (i) draw growth scenario and competitor entry; (ii) for each of 48 months draw $D_m \sim \mathcal{N}(\mu_m, \sigma_m^2)$ (clipped at zero); (iii) compute monthly profit using the capacity-constrained optimal Q_m^* ; (iv) aggregate to annual totals and discount. Table 7 reports the full risk profile.

Table 7: Monte Carlo risk measures ($N = 50000$, all values in £M except $\mathbb{P}(\text{NPV} < 0)$).

Measure	S1	S2	S3
$\mathbb{E}[\text{NPV}]$	6.58	46.54	21.50
Std. deviation	6.71	4.35	10.51
$\text{VaR}_{5\%}$	-1.10	41.69	6.10
$\text{CVaR}_{5\%}$	-1.97	41.19	5.01
$\mathbb{P}(\text{NPV} < 0)$	13.8%	0.0%	0.0%
10th percentile	-0.38	42.12	7.23
90th percentile	16.37	53.06	31.74

Strategy 2 is superior on *every* risk dimension simultaneously—it has the highest $\mathbb{E}[\text{NPV}]$, the *lowest* standard deviation, the highest VaR and CVaR, and zero probability of a net loss. This is a **stochastic dominance** result: Strategy 2 first-order stochastically dominates S1 and S3 (visible in the CDF plot, Figure 2).

Strategy 1 has a 13.8% probability of a net loss—primarily driven by the low-growth scenario where the £54M capex is not recovered within 4 years. Its NPV distribution has a wide spread (std. £6.71M) due to the combination of demand uncertainty and scenario uncertainty.

Strategy 3 has zero probability of loss and moderate expected return, but its NPV distribution is bimodal (std. £10.51M): the high-growth branch yields £12.4M (H, no comp.) or £7.0M (H+comp.) while the low-growth branch yields £30.8M—reflecting that the real option is actually more valuable under *low* growth (where flexibility investment is cheap and high-capacity expansion is avoided).

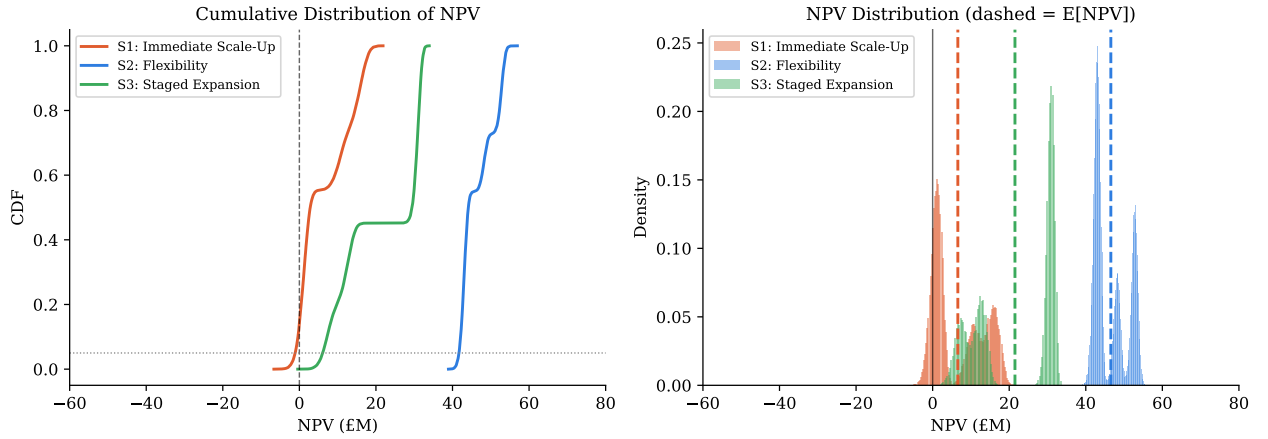


Figure 2: Left: Empirical CDF of simulated NPV for each strategy. Right: NPV density overlays (dashed lines = $\mathbb{E}[\text{NPV}]$). S2 distribution is tightly concentrated above £40M; S1 has significant left-tail mass below zero.

5.2 Sensitivity Analysis

Figure 3 shows $\mathbb{E}[\text{NPV}]$ as each parameter is varied individually while holding others at their base values.

Sensitivity of E[NPV] to Key Parameters

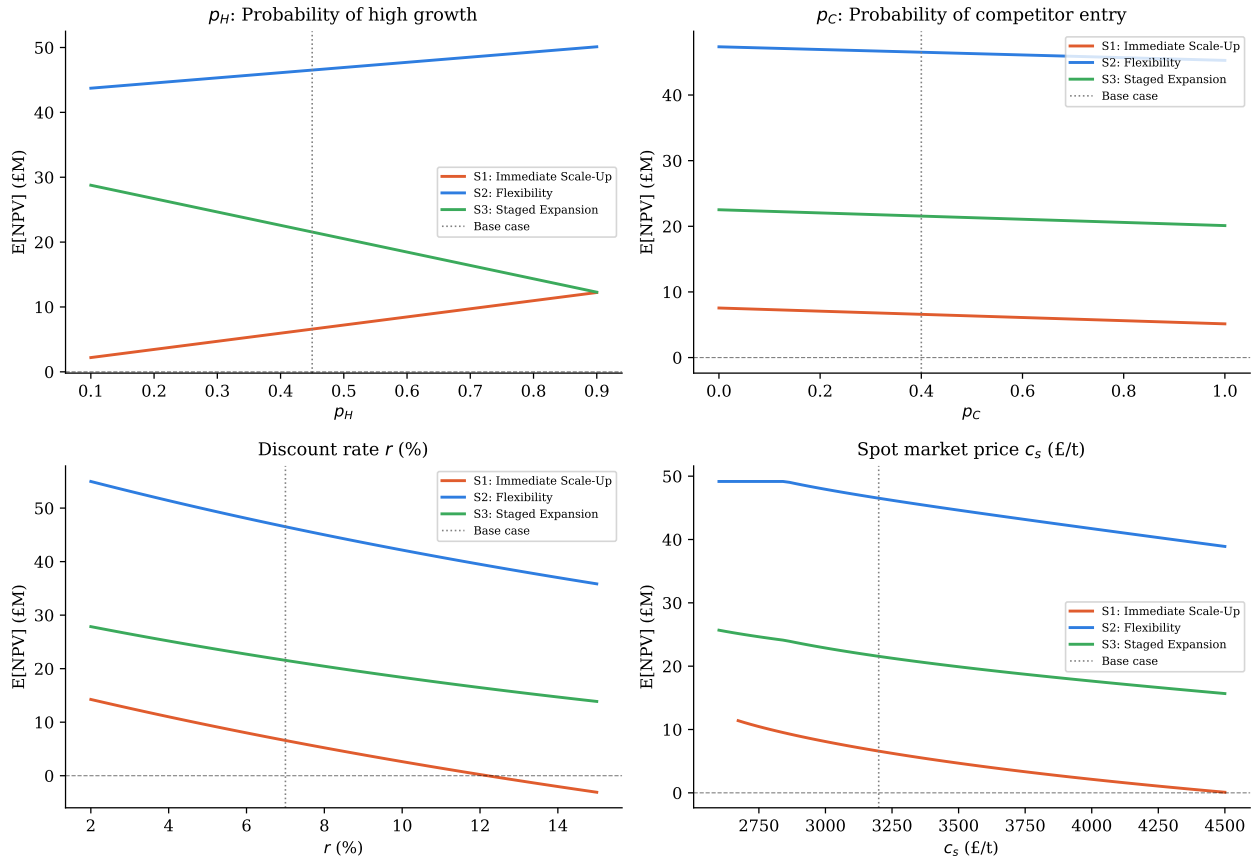


Figure 3: Sensitivity of $\mathbb{E}[\text{NPV}]$ to: growth probability p_H (top-left), competitor entry probability p_C (top-right), discount rate r (bottom-left), and spot price c_s (bottom-right). Dotted vertical line = base-case value.

Key findings:

- **p_H (growth probability):** S2 dominates across the entire range $p_H \in [0.1, 0.9]$ with no crossover. S3 overtakes S1 when $p_H \lesssim 0.65$, consistent with S3's relatively stronger performance under low growth. S1 is only competitive at very high p_H but still never surpasses S2.
- **p_C (competitor probability):** All strategies decline with p_C , but S2 remains far ahead. S1 declines faster because its large capacity expansion is most exposed to demand reduction.
- **Discount rate r :** Higher r penalises all strategies (longer-horizon cash flows are discounted more heavily), but affects S1 most because its capex is largest and front-loaded, while S2's lower capex means less unrecovered investment risk.
- **Spot market price c_s :** As c_s rises, all strategies benefit (higher spot prices increase revenue relative to procurement cost—but the model holds p fixed). S2's advantage actually *widens* at higher c_s , because its pre-negotiated contingent contracts keep $c_s^{S2} = c_s - £300$ while the others pay the full market rate.
- **Robustness:** No parameter variation causes the strategy ranking to change. S2's dominance is not an artefact of particular parameter choices—it is structurally robust.

6 Recommendations

6.1 Strategic Recommendation: Adopt Strategy 2

GreenCo should adopt Strategy 2 (Flexibility and Risk Mitigation). This recommendation is robust across all metrics: $\mathbb{E}[\text{NPV}] = £46.5\text{M}$ (vs $£6.6\text{M}$ for S1 and $£21.6\text{M}$ for S3), zero probability of net loss, lowest NPV volatility, and highest VaR/CVaR. Sensitivity analysis confirms S2 dominates across the full range of plausible parameter values with no breakeven.

The fundamental reason is economic: the newsvendor cost structure (overage cost $£2800/\text{t}$ *four times* the underage cost $£700/\text{t}$) means GreenCo should produce conservatively and rely on spot markets for shortfalls.

Adding large electrolyser capacity (S1/S3) exacerbates the overcapacity problem under low growth while providing only marginal benefit under high growth. Strategy 2 instead *reduces demand uncertainty*, which simultaneously reduces both spot procurement waste (lower ELS) and inventory waste (lower ELOI)—addressing the root cause of profit variability rather than throwing capacity at it.

Under high growth, S2's systematic reliance on spot procurement from Year 3 is a genuine limitation. However, even in this scenario S2 yields NPV = £52.75M (no comp.) or £48.15M (with comp.), both far exceeding S1 and S3. If high growth materialises and spot market prices rise significantly, S2's pre-negotiated contingent contracts at £2900/t provide important downside protection.

6.2 Operational Recommendations

1. **Production policy:** Implement the newsvendor-optimal schedule $Q_t^* = \min(\mu_t + z^* \sigma_t, \bar{Q})$, reviewed annually as demand parameters update. Under Strategy 2, use $z^{*,S2} = \Phi^{-1}(0.125) = -1.150$ (reduced c_s lowers the underage penalty, justifying further underproduction). *Avoid planning at mean demand*—this ignores the asymmetric cost structure and the flaw of averages.
2. **Refuelling station: add bays immediately.** The current M/M/3 configuration violates the 5-minute SLA at baseline ($W_q = 5.33$ min) and deteriorates rapidly under demand growth. *This is independent of the investment strategy chosen.* Two additional bays ($s = 5$) are required for compliance across the 4-year horizon. With $s = 5$: $W_q = 0.24$ min at baseline, rising to 1.27 min under high growth by Y4. A single additional bay ($s = 4$) is insufficient under high growth (fails by Y4 at $\lambda = 14.64/\text{hr} > \lambda_{s=4}^{\max} = 14.44/\text{hr}$). The cost of non-compliance (contract breach) likely far exceeds the incremental bay installation cost.
3. **Demand monitoring:** Under Strategy 2, the firm should monitor realised demand growth closely. If high growth materialises earlier than expected, activating an option to expand capacity (analogous to S3) at a premium cost may become worthwhile from Year 2 onward.

6.3 Limitations and Caveats

- **Demand independence:** Monthly demands are assumed i.i.d. within each year. Serial correlation (e.g. contract ramp-ups) would alter the newsvendor solution and widen the simulated NPV distribution.
- **Risk neutrality:** The analysis maximises $E[\text{NPV}]$. A risk-averse decision-maker would assign additional weight to downside outcomes; this would strengthen the case for S2 (already has zero loss probability) and weaken S1 further.
- **Flaw of averages:** Computing profits at expected demand μ_t rather than simulating monthly realisations underestimates costs due to the nonlinear capacity-demand interaction (Jensen's inequality). The Monte Carlo simulation explicitly addresses this; the analytical $E[\text{NPV}]$ figures are close to but slightly above the simulated means (e.g. S2: £46.52M analytical vs £46.54M simulated).
- **Strategy 2 spot exposure under high growth:** From Year 3 onward under high growth, Strategy 2 relies systematically on spot procurement. If spot market liquidity is limited or prices spike, the pre-negotiated contingent contracts at £2900/t are critical. If these contracts cannot be honoured at scale, S2's advantage would narrow.
- **Normal demand assumption:** Negative demand realisations are theoretically possible but negligible in practice ($> 6\sigma$ from mean). Demand is truncated at zero in simulation.
- **Queueing model assumptions:** Steady-state M/M/s assumes stationary Poisson arrivals. In practice, HGV arrivals may exhibit time-of-day patterns; a time-varying $\lambda(t)$ model would give a more accurate picture of peak congestion.

References

- [1] E. Gungor, *3E3 Modelling Risk: Lecture Notes*, University of Cambridge, Lent 2026.